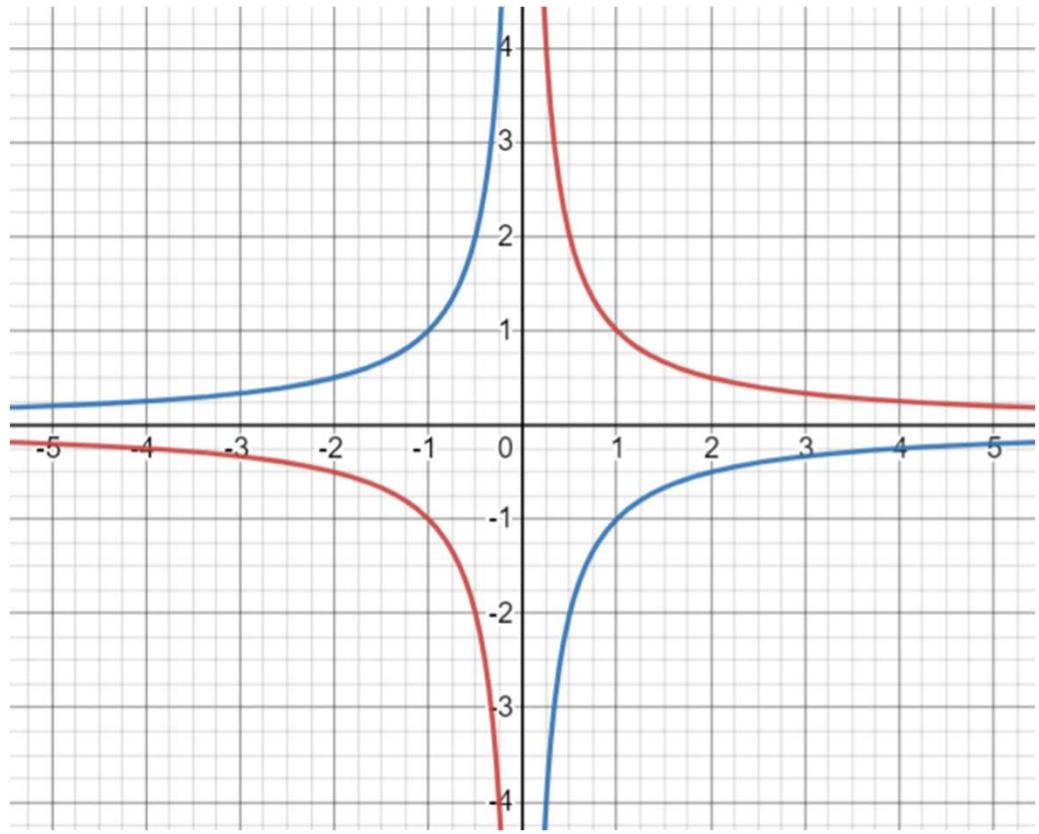
Translations and transformations of parent rational function

Multiplying by a negative coefficient:

$$y = \frac{1}{x}$$

2

$$y = -\frac{1}{x}$$



Translations of parent rational function

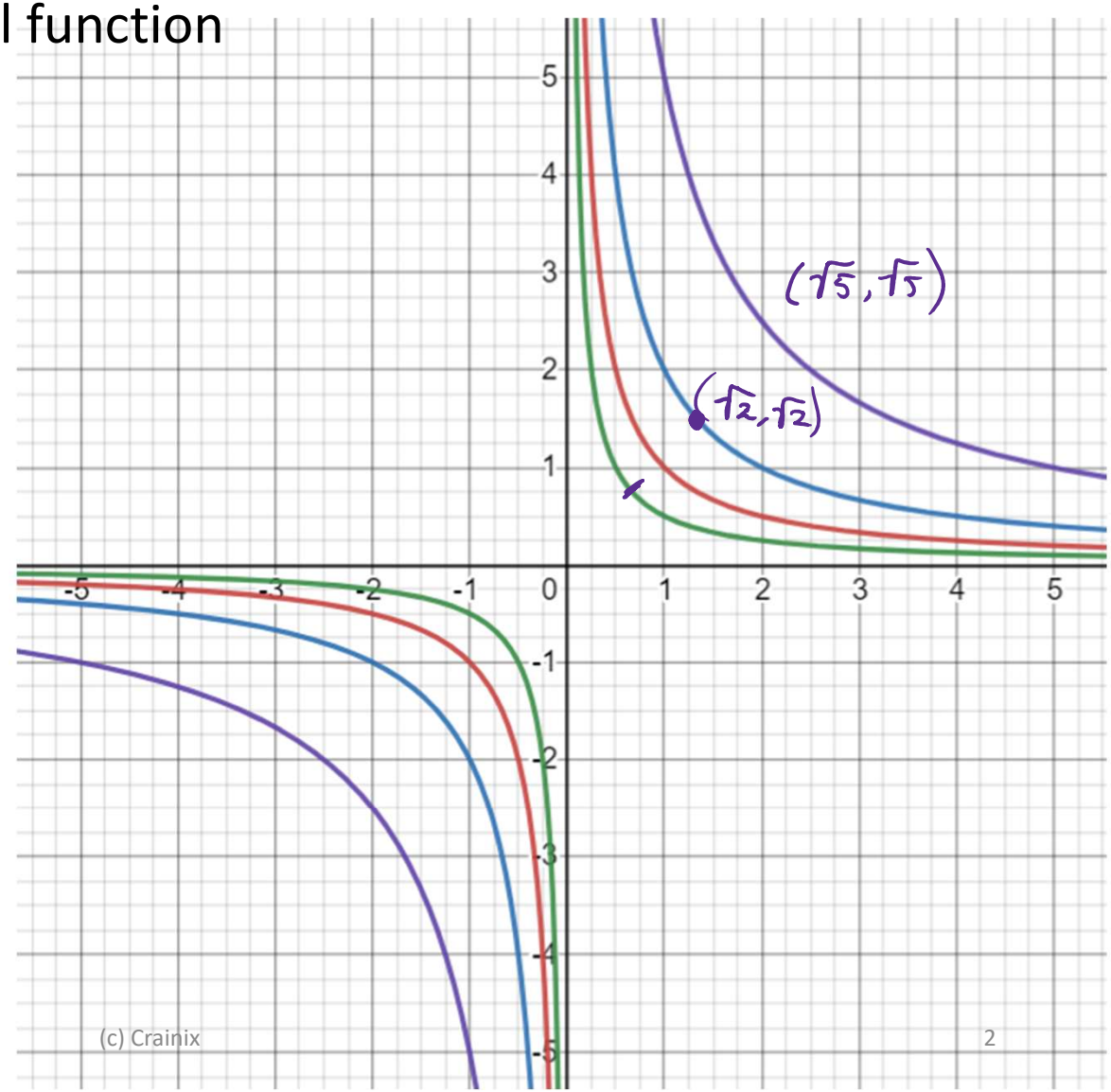
Multiplying by a coefficient:

1 $y = \frac{1}{x}$

2 $y = \frac{2}{x}$ $\quad y \cdot x = 2$

3 $y = \frac{1}{2x}$ $\quad yx = \frac{1}{2} \text{ or } \frac{1}{\sqrt{2}}$

4 $y = \frac{5}{x}$ $\quad y \cdot x = 5$

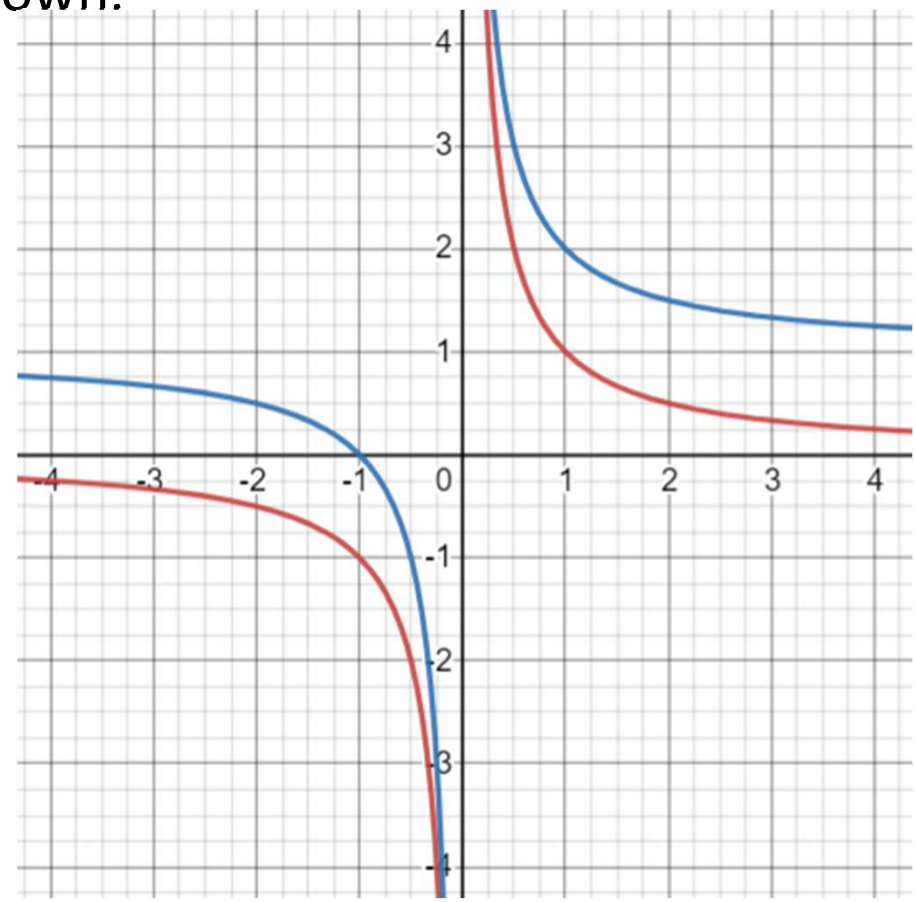


Translations of parent rational function

Adding a constant shifts it up or down.

$$y = \frac{1}{x}$$

$$y = \frac{1}{x} + 1$$



Translations of parent rational function

Inside the function with the x, it goes right when – and left when +

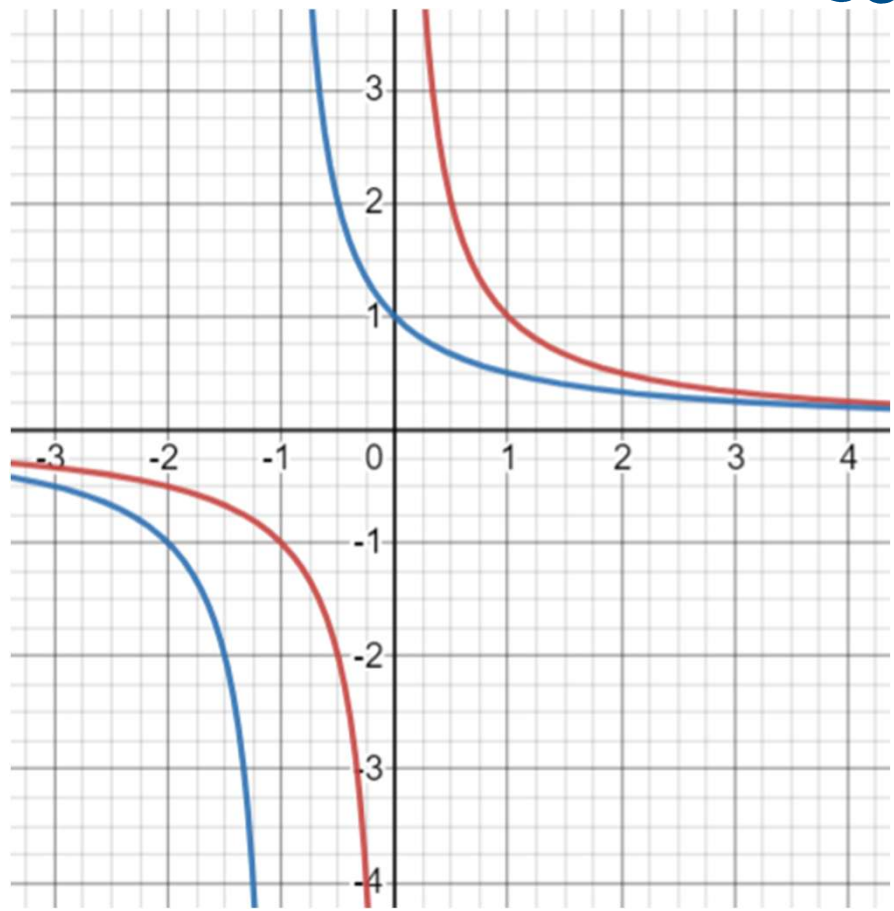
$1 + \frac{1}{1000000}$

1 $y = \frac{1}{x}$

2 $y = \frac{1}{(x+1)}$ $x \neq -1$

$x + 1 \neq 0$

$x \neq -1$

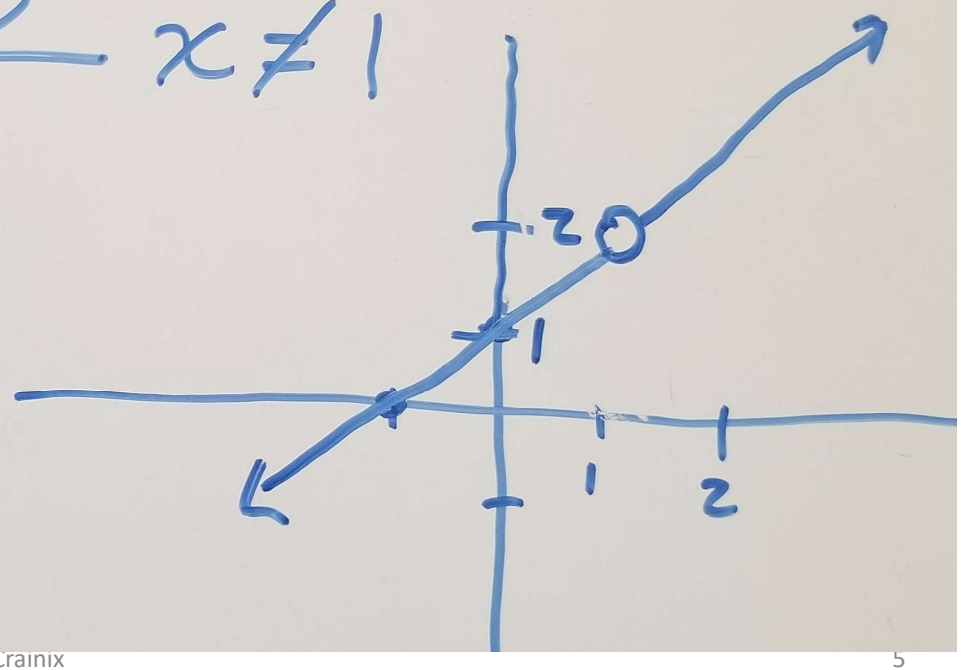


10000001

Polynomials over polynomials: if it factors to a line, there is a hole or a **point discontinuity** or a **removable singularity** at the place where x would make the function divide by zero.

$$y = \frac{(x+1)(x-1)}{(x-1)} \quad x \neq 1$$

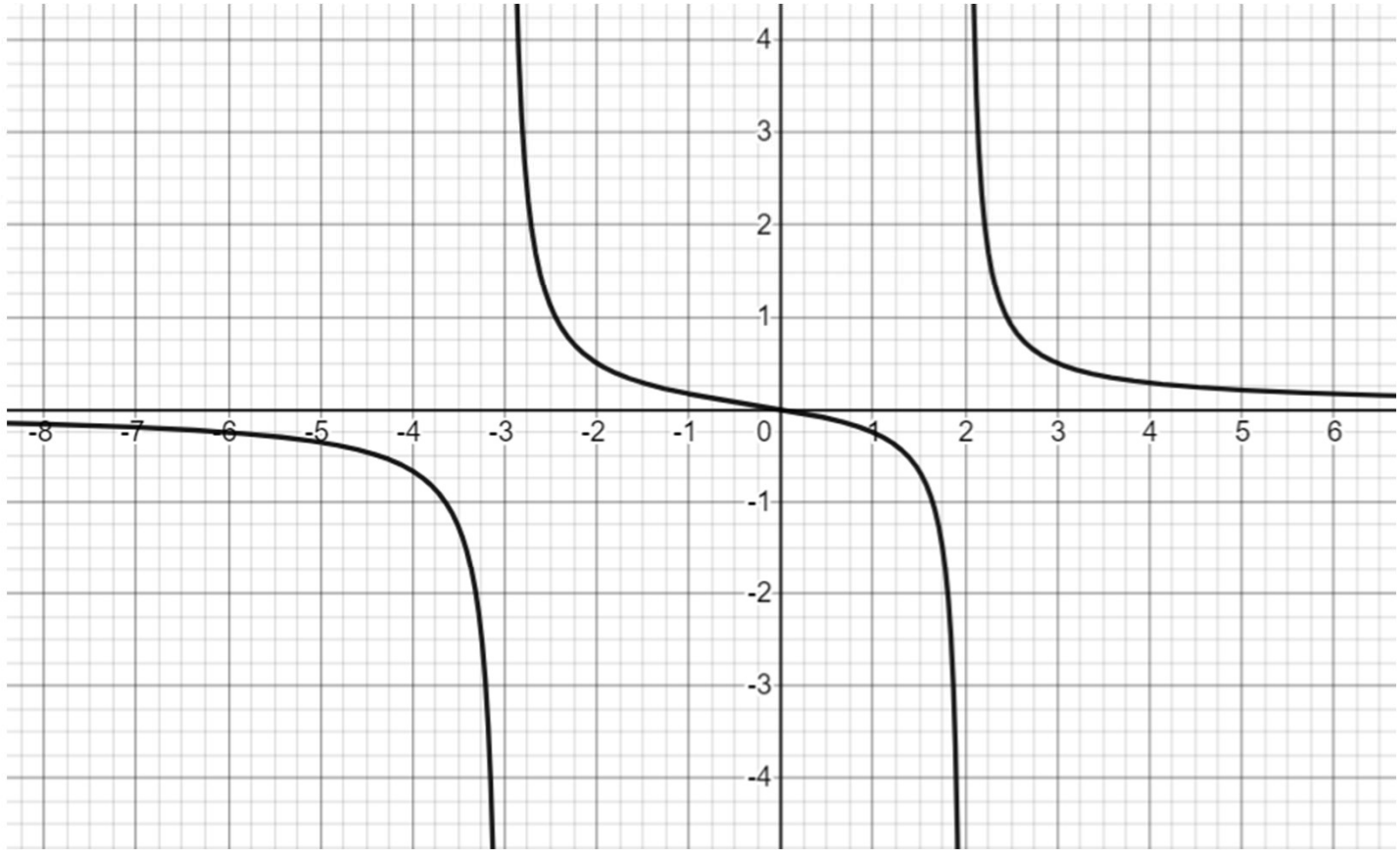
$$y = x+1 \quad x \neq 1$$



Polynomials over polynomials:

Vertical asymptotes occur at the x value where the denominator would be zero when the numerator isn't zero.

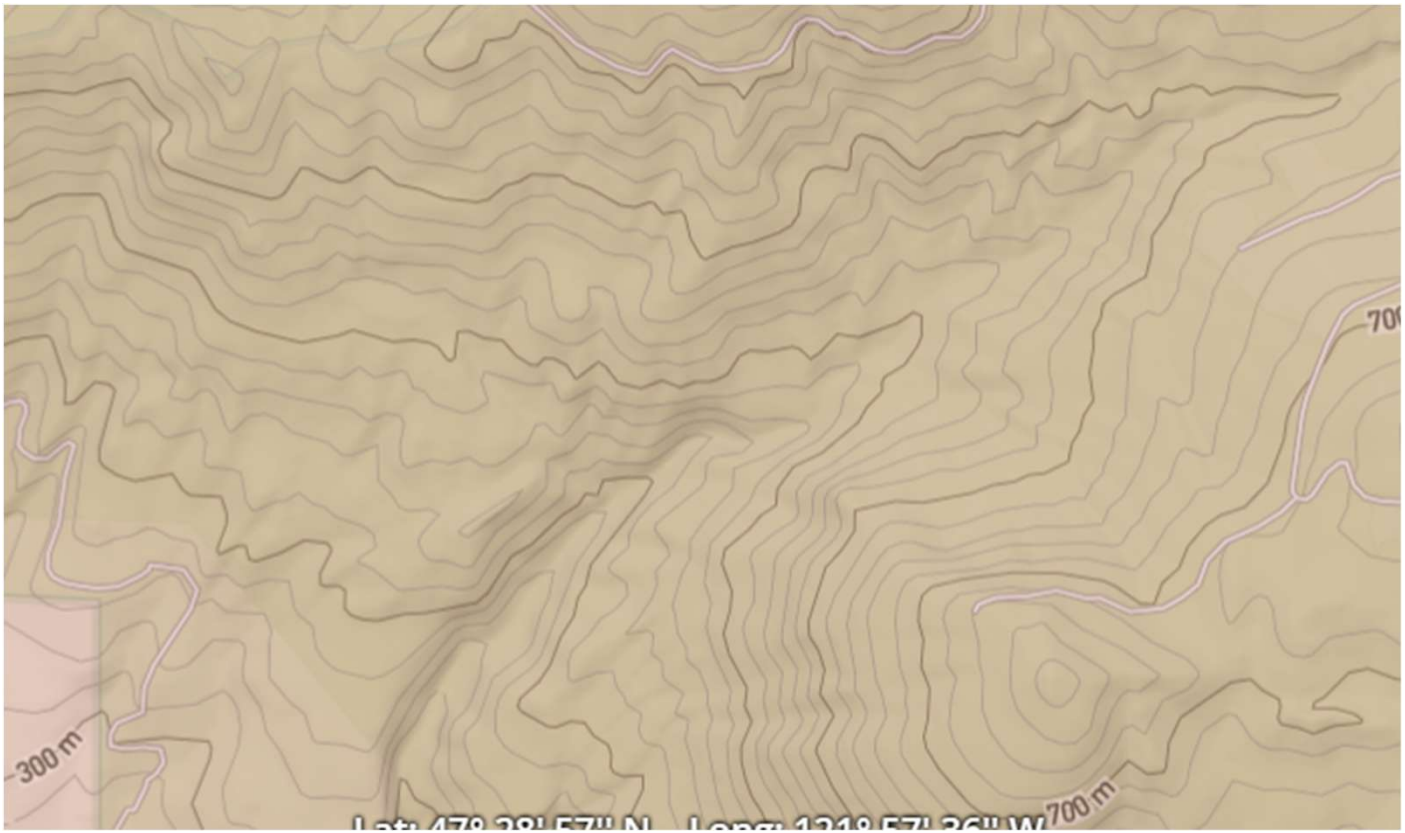
$$y = \frac{x}{(x+3)(x-2)}$$



Topographical maps can sometimes be modeled using rational functions.

Here is a topological map of Tiger Mountain, Washington.

Get Maps | topoView (usgs.gov)



Horizontal asymptotes for Precalc classes: Slant asymptotes aren't standard math terminology.

$$y = \frac{ax^{hi} + \cdots}{bx^{lo} + \cdots} \qquad \frac{2x^5+1}{3x^4+6}$$

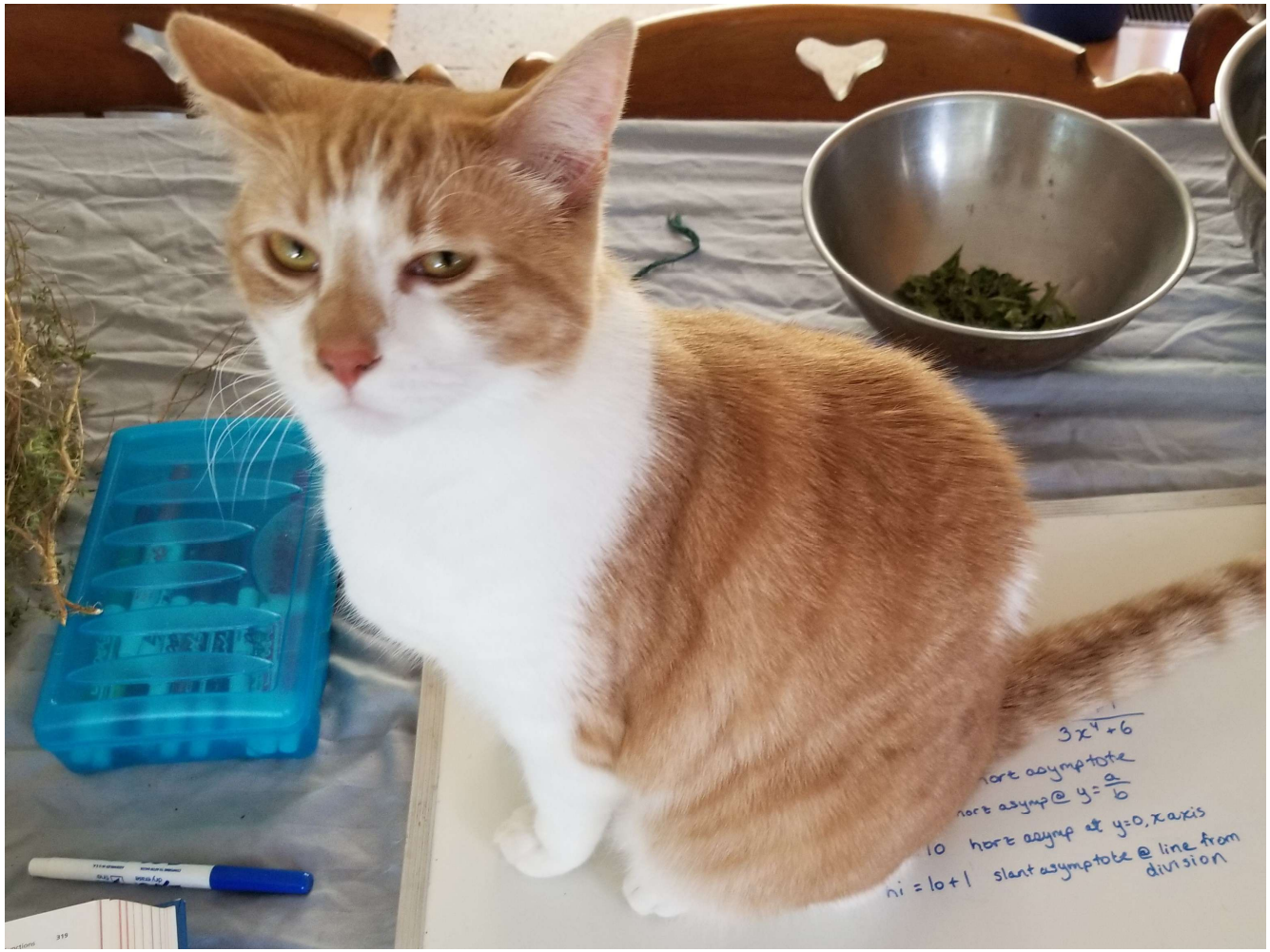
$hi > lo$ no horz asymptote

$hi = lo$ horz asymp @ $y = \frac{a}{b}$

$hi < lo$ horz asymp at $y=0$, x axis

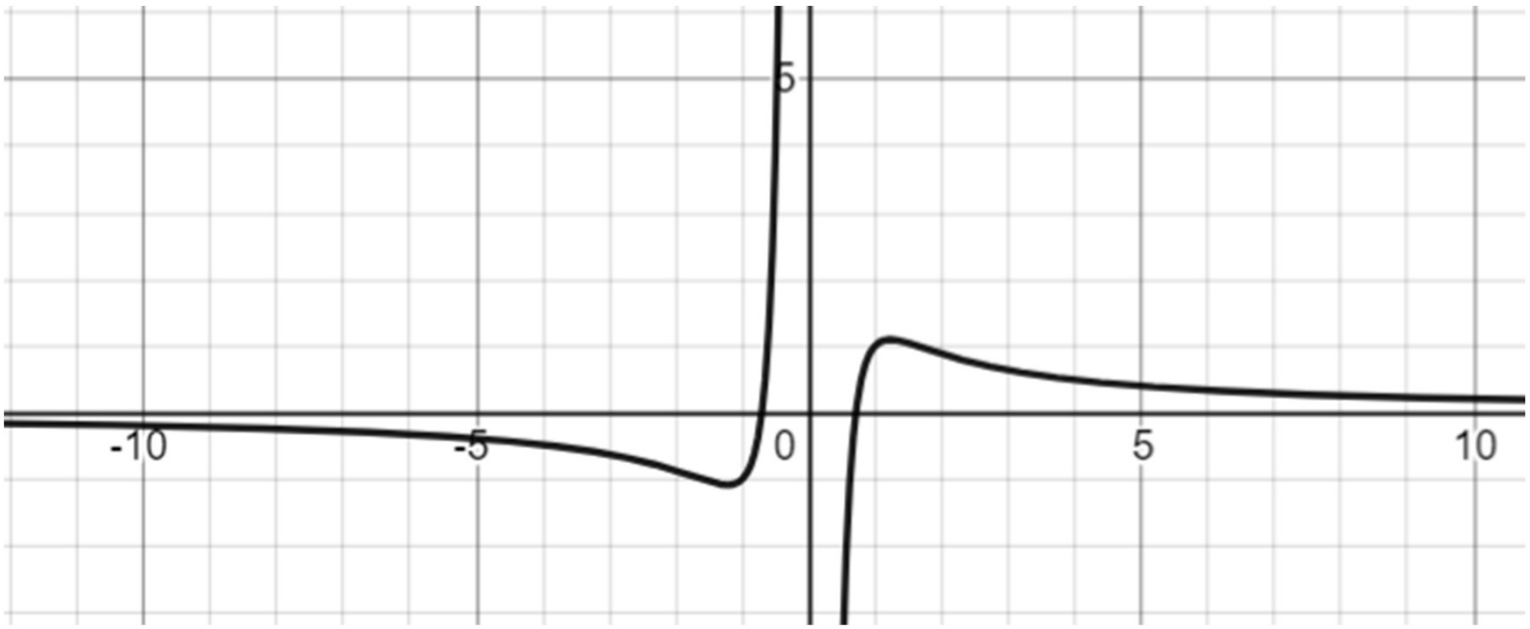
$hi = lo + 1$ slant asymptote @ line from division

The TA doesn’t think we should bother with this, but we can do it!



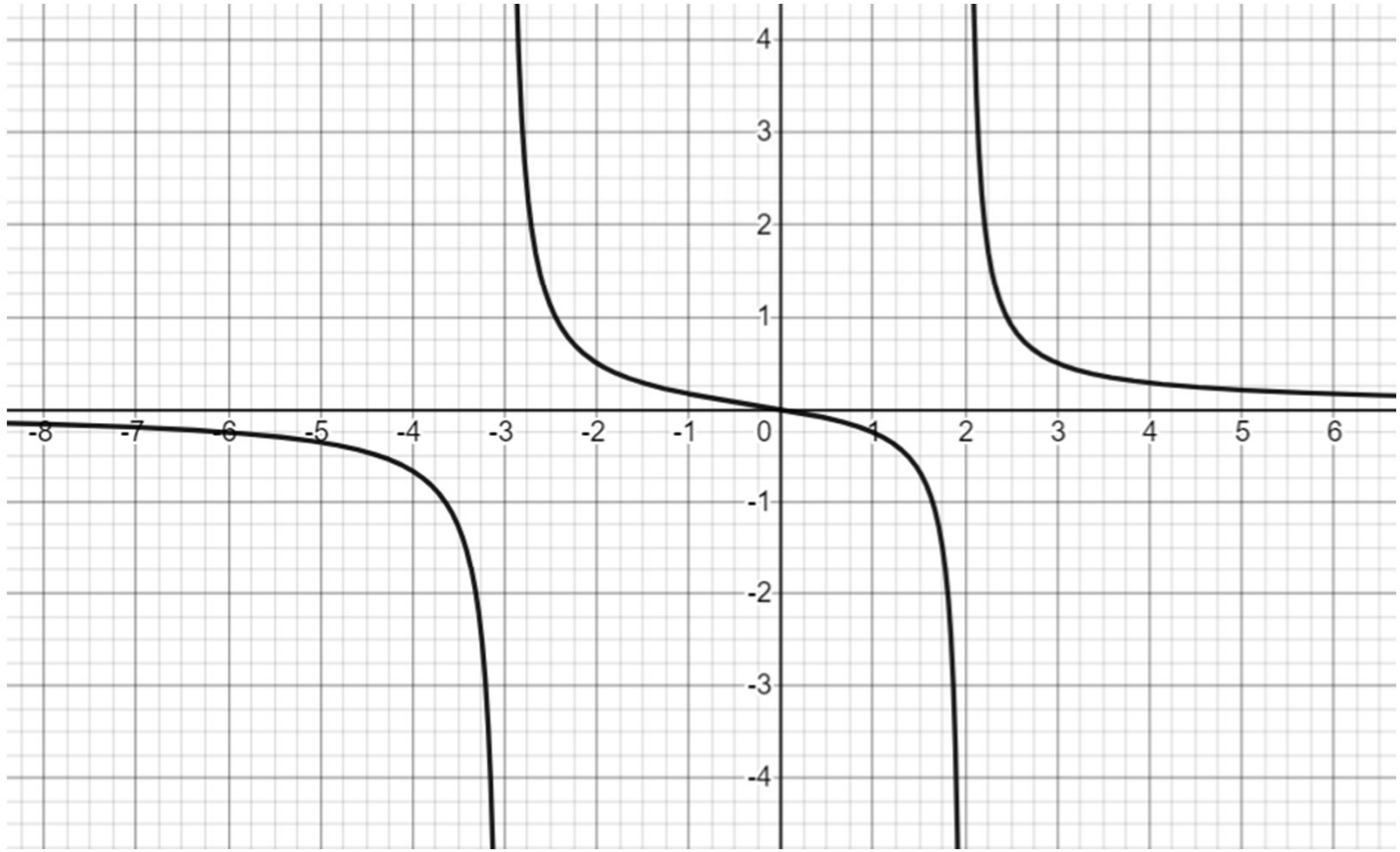
Horizontal asymptotes: Top highest power less than bottom highest power gives a horizontal asymptote at x axis or y=0

$$y = \frac{2x^2 - 1}{x^3}$$



Horizontal asymptotes: Top highest power less than bottom highest power gives a horizontal asymptote at x axis or y=0

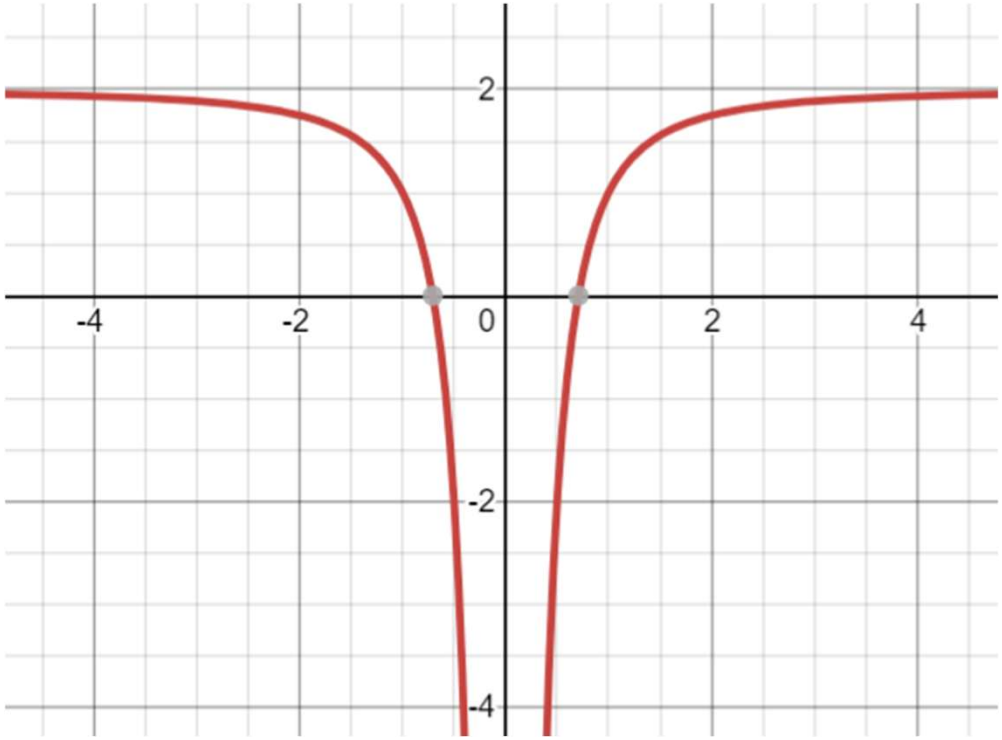
$$y = \frac{x}{(x+3)(x-2)}$$



Horizontal asymptotes: same highest power give a horizontal asymptote at y=a/b

4/2 is 2 so the horizontal asymptote is at 2

$$\frac{4x^2 - 1}{2x^2}$$



Slant asymptote (not a mathematical term) at the line that you get when top power is one more than bottom power and you do the division with a remainder. You may need for testing.

$$y = x$$

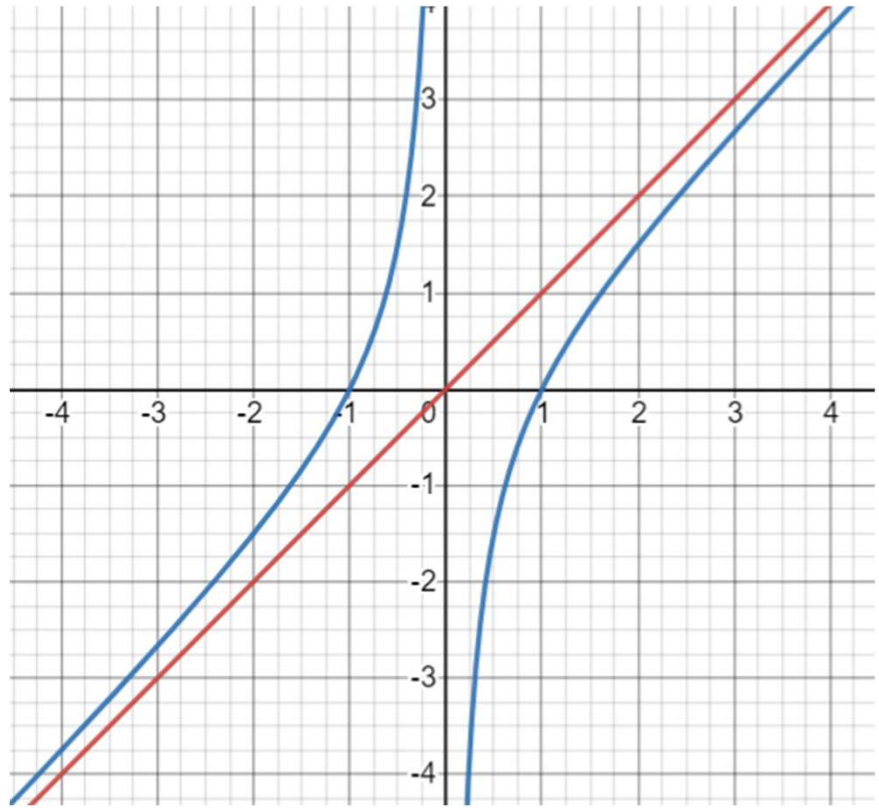
2

$$y = \frac{x^2 - 1}{x}$$

$$\frac{x^2-1}{x} = 1x + \frac{-1}{x}$$

$$x\left(x - \frac{1}{x}\right)$$

$$x^2 - 1$$

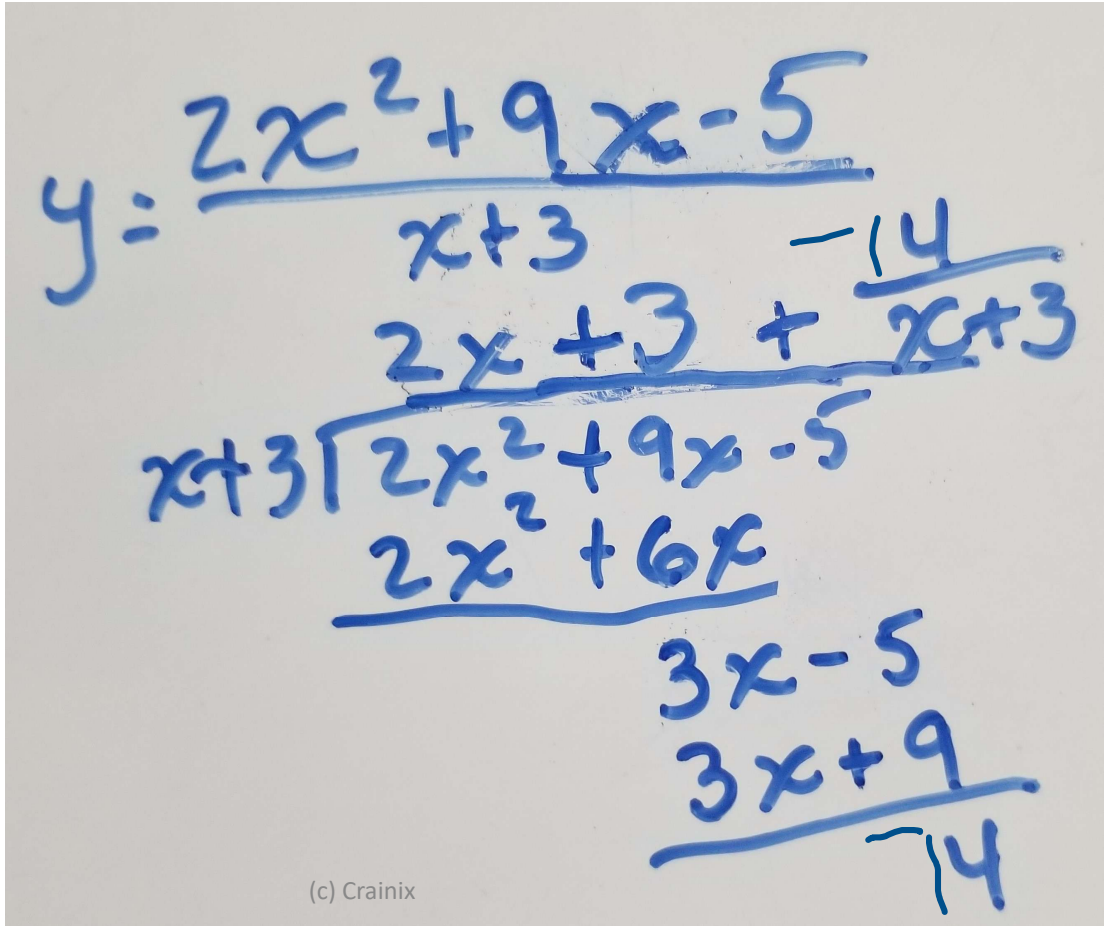


Slant asymptote at the line that you get when top power is one more than bottom power and you do the division with a remainder.

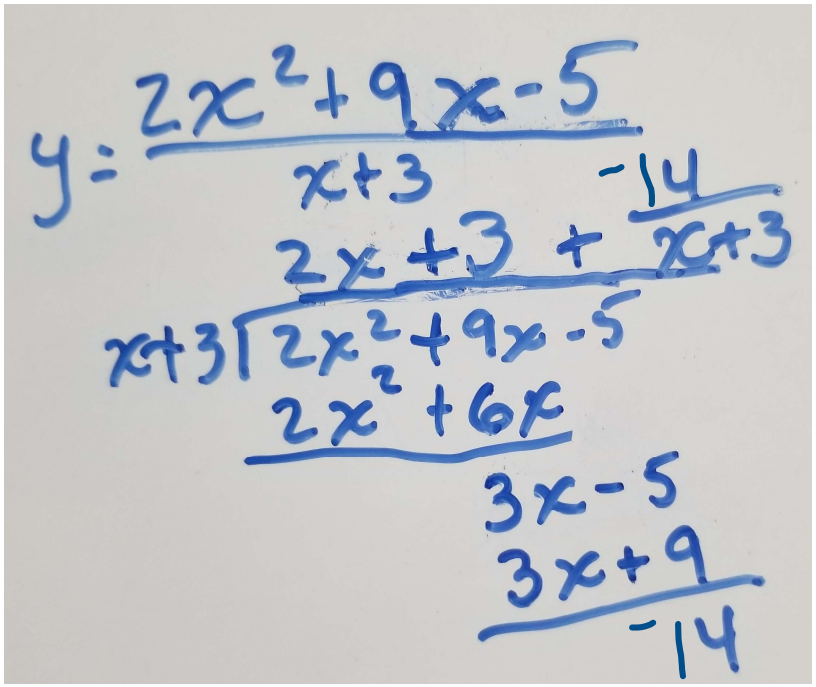
$$y = \frac{\left(2x^2 + 9x - 5\right)}{x + 3}$$

2

$$y = 2x + 3$$



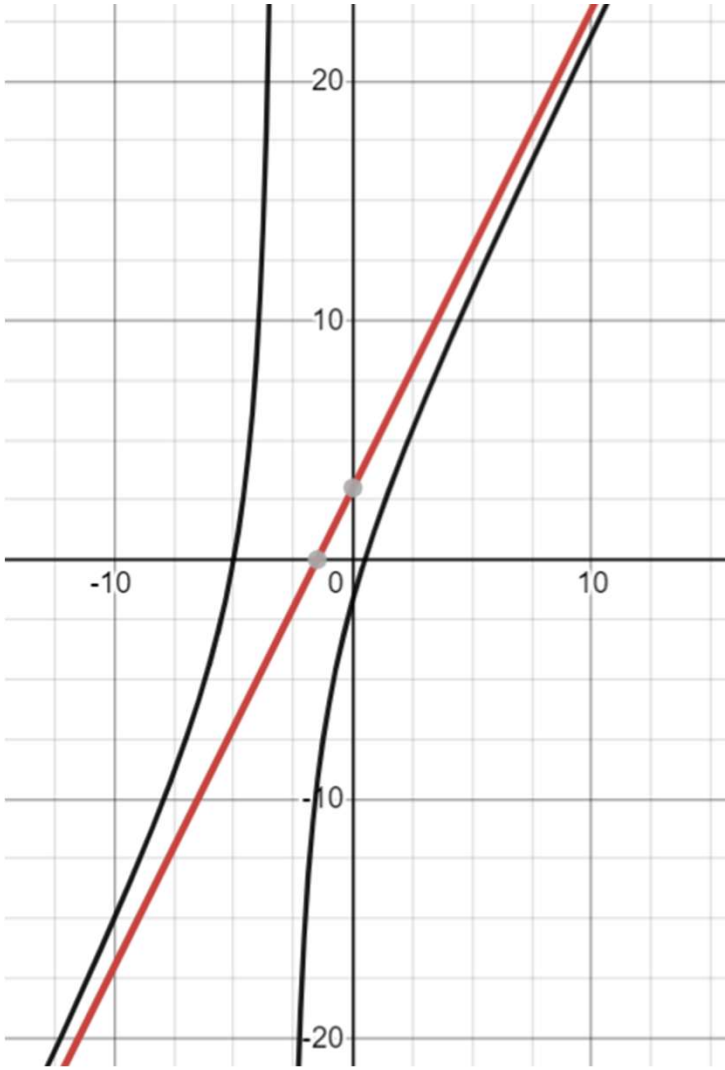
Slant asymptote at the line that you get when top power is one more than bottom power and you do the division with a remainder.



$y = \frac{(2x^2+9x-5)}{x+3}$

2

$y = 2x+3$



Horizontal asymptotes

$$y = \frac{a x^{hi} + \cdots}{b x^{lo} + \cdots} \qquad \frac{2x^5 + 1}{3x^4 + 6}$$

$hi > lo$ no horz asymptote

$hi = lo$ horz asymp @ $y = \frac{a}{b}$

$hi < lo$ horz asymp at $y = 0$, x axis

$hi = lo + 1$ slant asymptote @ line from division